

- Bazı Önemli Eşitsizlikler -

Cauchy-Schwartz Eşitsizliği: $\vec{u}, \vec{v} \in \mathbb{R}^n$ için $|\langle \vec{u}, \vec{v} \rangle| \leq \|\vec{u}\| \cdot \|\vec{v}\|$ olur.

Kanıt: $\vec{u} \neq \vec{0}$ ve $\vec{v} \neq \vec{0}$ ($\vec{u} = \vec{0}$ veya $\vec{v} = \vec{0}$ ise eşitsizlik okulu)

$$\langle \vec{u}, \vec{v} \rangle = \|\vec{u}\| \cdot \|\vec{v}\| \cdot \cos \theta$$

$$\Rightarrow \cos \theta = \frac{\langle \vec{u}, \vec{v} \rangle}{\|\vec{u}\| \cdot \|\vec{v}\|}$$

$$\Rightarrow -1 \leq \frac{\langle \vec{u}, \vec{v} \rangle}{\|\vec{u}\| \cdot \|\vec{v}\|} \leq 1 \Rightarrow -\|\vec{u}\| \cdot \|\vec{v}\| \leq \langle \vec{u}, \vec{v} \rangle \leq \|\vec{u}\| \cdot \|\vec{v}\|$$
$$\Rightarrow |\langle \vec{u}, \vec{v} \rangle| \leq \|\vec{u}\| \cdot \|\vec{v}\| \quad \#$$

Üçgen Eşitsizliği: $\vec{u}, \vec{v} \in \mathbb{R}^n$ için $\|\vec{u} + \vec{v}\| \leq \|\vec{u}\| + \|\vec{v}\|$ dir.

Kanıt:

$$\begin{aligned}\|\vec{u} + \vec{v}\|^2 &= \langle \vec{u} + \vec{v}, \vec{u} + \vec{v} \rangle \\&= \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle \\&= \|\vec{u}\|^2 + 2 \cdot \langle \vec{u}, \vec{v} \rangle + \|\vec{v}\|^2 \\&\leq \|\vec{u}\|^2 + 2 |\langle \vec{u}, \vec{v} \rangle| + \|\vec{v}\|^2 \\&\leq \|\vec{u}\|^2 + 2 \cdot \|\vec{u}\| \cdot \|\vec{v}\| + \|\vec{v}\|^2 \\&= (\|\vec{u}\| + \|\vec{v}\|)^2\end{aligned}$$

$$\Rightarrow \|\vec{u} + \vec{v}\| \leq \|\vec{u}\| + \|\vec{v}\| \quad \#$$

Paralelkenar Eşitliği : $\vec{u}, \vec{v} \in \mathbb{R}^n$ için

olur.

$$\|\vec{u} + \vec{v}\|^2 + \|\vec{u} - \vec{v}\|^2 = 2(\|\vec{u}\|^2 + \|\vec{v}\|^2)$$

Kanıt :

$$\begin{aligned}\|\vec{u} + \vec{v}\|^2 + \|\vec{u} - \vec{v}\|^2 &= \langle \vec{u} + \vec{v}, \vec{u} + \vec{v} \rangle + \langle \vec{u} - \vec{v}, \vec{u} - \vec{v} \rangle \\ &= 2\langle \vec{u}, \vec{u} \rangle + 2\langle \vec{v}, \vec{v} \rangle \\ &= 2(\|\vec{u}\|^2 + \|\vec{v}\|^2)\end{aligned}$$

Polarizasyon Eşitliği : $\vec{u}, \vec{v} \in \mathbb{R}^n$ olmak üzere

$$\langle \vec{u}, \vec{v} \rangle = \frac{1}{4} \cdot \|\vec{u} + \vec{v}\|^2 - \frac{1}{4} \|\vec{u} - \vec{v}\|^2$$

olur.

Kanıt :

$$\begin{aligned} \frac{1}{4} \cdot \|\vec{u} + \vec{v}\|^2 &= \frac{1}{4} \cdot \left(\|\vec{u}\|^2 + 2\langle \vec{u}, \vec{v} \rangle + \|\vec{v}\|^2 \right) \\ + / \frac{1}{4} \cdot \|\vec{u} - \vec{v}\|^2 &= \frac{1}{4} \cdot \left(\|\vec{u}\|^2 - 2\langle \vec{u}, \vec{v} \rangle + \|\vec{v}\|^2 \right) \\ \hline &= \langle \vec{u}, \vec{v} \rangle \end{aligned}$$